Deployment Architecture and Environment Specification

For Oromo Research Associations (ORA) Platform

1. Overview

The Oromo Research Associations (ORA) Platform is a scalable, secure, and high-performance digital platform designed to support research publication, academic collaboration, and knowledge dissemination. The platform is composed of six functional modules:

1. Journal Management
2. E-Books
3. Library Management
4. Research Repository
5. Oromo Wikipedia
6. Researcher Networks

The system is built using a micro services architecture, with all services deployed as containerized workloads orchestrated. The architecture prioritizes high availability, fault tolerance, security, and efficient handling of large documents and media files

2. Requirements Overview

2.1 Functional Requirements

The ORA Platform must support:

- Journal article submission, review, and publication
- Hosting and distribution of e-books and digital resources
- Library cataloging and resource management
- Research paper and dataset archiving
- Collaborative knowledge creation (Oromo Wikipedia)
- Researcher profiles, networking, and collaboration tools

2.2 Non-Functional Requirements

- **Availability:** ≥ 99.9% uptime
- **Scalability:** Horizontal scaling of microservices
- **Performance:** Low-latency APIs and high-throughput media handling
- **Security:** Encrypted communication, VPN access, role-based authorization
- **Data Protection:** Replication, backups, and logical isolation
- **Cost Efficiency:** Optimized VM sizing and resource utilization

3. Proposed Deployment Architecture

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The ORA Platform is deployed as a distributed microservices system orchestrated, using a combination of stateless and state full services deployed across multiple virtual machines.

Core Infrastructure Components

- Docker: Container runtime
- NGINX: Reverse proxy, TLS termination, and load balancing
- PostgreSQL: Relational database
- MinIO: Object storage for documents and media
- Redis: Caching and session management
- Prometheus, Grafana, Loki: Monitoring and logging
- VPN Gateway: Secure internal access

4. Design Principles and Justification – Infrastructure

4.1 Micro services Architecture

Each ORA module is implemented as a **separate micro service**, independently deployable and scalable. This enables:

- Faster development cycles
- Fault isolation
- Independent scaling of heavy-use modules (e.g., Repository, E-Books)

Infrastructure services (PostgreSQL, MinIO, Redis, monitoring) are logically and physically separated from application services, simplifying maintenance and scaling.

4.2 High Availability & Fault Tolerance

The system is designed to remain operational despite individual component failures:

- **Nomad Cluster:**
  - 3 Nomad servers (1 leader, 2 followers)
  - Automatic leader election and failover
- **Nomad Clients:**
  - Multiple clients hosting replicated microservices
  - Services distributed across nodes to avoid single-point failure
- **PostgreSQL:**
  - Active-standby replication
  - Automatic failover

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- **MinIO:**
    - Distributed object storage with replication
    - High durability for research documents and media.
- **NGINX:**
    - Active-passive configuration
    - Seamless traffic failover

4.3 Performance & Scalability

- Horizontal scaling of micro services via Nomad job replicas
- NGINX load-balances traffic across service instances
- Redis reduces database load through caching
- MinIO handles large file uploads/downloads independently of the database
- Supports high concurrency for document and media-intensive workloads

4.4 Security Architecture

Security is enforced at multiple layers:

- **TLS/SSL encryption** for all external and internal traffic
- **JWT-based authentication**
- **Role-Based Access Control (RBAC)**
- **VPN access** for internal editors, administrators, and researchers
- **Service isolation** using Nomad job boundaries
- Reduced blast radius in case of service compromise

4.5 Backup & Disaster Recovery

- **PostgreSQL:**
    - Continuous replication
    - Scheduled full and incremental backups
- **MinIO:**
    - Object replication

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- External backup to offline or cloud storage

- **Backup Node:**
  - Dedicated backup pipelines
  - Optional long-term archival storage

5. Network Bandwidth Considerations

The ORA Platform supports large documents, scanned manuscripts, PDFs, datasets, audio, and video content. Expected usage includes thousands of concurrent users accessing and transferring files ranging from 5 MB to 500 MB.

To ensure performance and reliability:

- 5 Gbps external internet link is recommended
- Adequate bandwidth headroom for future growth
- Stable encrypted transfers for sensitive academic data

6. ORA Platform Micro services Breakdown

6.1 Journal Management Service

- Article submission and review
- Editorial workflows
- PDF and metadata storage
- Indexing and search APIs

6.2 E-Books Service

- Digital book upload and management
- Controlled access and downloads
- Metadata and categorization

6.3 Library Management Service

- Catalog management
- Resource tracking
- Integration with digital collections

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6.4 Repository Management Service

- Research papers and datasets
- Versioning and access control
- Long-term archival storage

6.5 Oromo Wikipedia Service

- Collaborative editing
- Moderation and revision history
- Public read access, restricted write access

6.6 Researcher Networks Service

- Researcher profiles
- Collaboration tools
- Messaging and notifications

7. VM & Resource Specification (Production)

Component Breakdown

Component	VM Count	vCPU	RAM	Disk	Role
Nomad Server	3	16	32 GB	500 GB	Nomad control plane
Nomad Client	5	32	96 GB	1.5 TB	Microservices hosting
PostgreSQL	2	32	128 GB	2 TB	Primary DB + standby
Redis	3	16	64 GB	500 GB	Cache & sessions
MinIO	2	16	128 GB	10 TB	Object storage
NGINX (HA)	2	16	32 GB	500 GB	TLS & load balancing
Monitoring Stack	1	16	32 GB	500 GB	Prometheus, Grafana, Loki
Backup Node	1	16	32 GB	External	Backups
Analytics / Reports	1	16	32 GB	500 GB	Reporting & insights
Total	**20 VMs**				

Special need resource needed, for Oromo Wikipedia Service

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Resource	*Value*
vCPU	*16*
RAM	*32 GB*
Disk	*500 GB*
Role	*Wikipedia frontend acceleration*

8. Risks & Mitigations

Risk	**Mitigation**
Node failure	Multi-node Nomad clusters
Data loss	Replication and external backups
High traffic	Horizontal service scaling
Security breach	Encryption, RBAC
Storage growth	Scalable MinIO storage

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